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C20-EE–503

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BOARD DIPLOMA EXAMINATION, (C-20)

MARCH/APRIL—2025

DEEE – FIFTH SEMESTER EXAMINATION

POWER SYSTEMS—III (SWITCH GEAR AND PROTECTION)

Time : 3 Hours] [*Total Marks* : 80

PART—A 3×10=30

Instructions : (1) Answer **all** questions.

(2) Each question carries **three** marks.

(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. State the phenomenon of arc.

2. Compare SF_6 and minimum oil circuit breaker in any three aspects.

3. Define *(a)* selectivity and *(b)* reliability.

4. List the types of faults in transformer.

5. State the need for field protection in an alternator.

6. List the advantages of buchholz relay.

7. List the causes of bus bar faults.

8. Write a short note on pilot wire protection.

9. Explain Facts in brief.

10. State the requirement of wirtricity.

PART—B

8×5=40

Instructions : (1) Answer **all** questions.

(2) Each question carries **eight** marks.

(3) Answers should be comprehensive and criterion for valuation is the content but not the length of the answer.

11. *(a)* Explain the construction and working of axial blast circuit breaker with neat diagram.

(OR)

(b) Explain the construction and working of minimum oil circuit breaker with neat diagram.

12. *(a)* Explain the construction and working principle of impedence relay.

(OR)

(b) Explain the working of induction type over current relay with diagram.

13. *(a)* Explain the scheme of differential protection of transformer.

(OR)

(b) Explain the scheme of protection against excessive heatintg of stator in alternator.

14. *(a)* Explain the protection of radial feeders by time graded relays.

(OR)

(b) Explain the different schemes of protections of bus bars.

15. *(a)* Explain the Protection of Transmission lines using Impedence relays.

(OR)

(b) Explain the protection of transmission lines using current grading.

PART—C

10×1=10

Instructions : (1) Answer the following question.

(2) The question carries **ten** marks.

(3) Answer should be comprehensive and the criterion for valuation is the content but not the length of the answer.

16. A generating station has two alternators of ratings 4000 kVA and 6000 kVA of percentage reactance 10% and 8% respectively connected from the common bus bars. The load is taken to the feeder through a 12000 kVA transformer of 5% reactance. What should be the short circuit kVA and approximate rating of circuit breaker if the fault occurs on the feeder?

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